TUTORIAL PLAN

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Branch	**Semester**	**Subject**	**Code**	**Session**
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Tutorial Sheet No. 2

Deadlocks: Introduction to deadlocks, Conditions for deadlock, Resource allocation graphs, Deadlock prevention and avoidance, Deadlock detection and recovery.

Q1. Suppose that the system is having 12 tape drives. At current time situation is:-

Process	Maximum Needs	Current Needs
P0	10	5
P1	4	2
P2	9	2

Answer the following questions using the banker's algorithm:

a. Illustrate that the system is in a safe state and what is safe sequence?
b. If process P2 request one for tape drive then safe sequence exist or not?

Q2. Consider the following snapshot of a system:

	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	2	0	0	1	4	2	1	2	3	3	2	1
P1	3	1	2	1	5	2	5	2				
P2	2	1	0	3	2	3	1	6				
P3	1	3	1	2	1	4	2	4				
P4	1	4	3	2	3	6	6	5				

Answer the following questions using the banker's algorithm:

a. Illustrate that the system is in a safe state by demonstrating an order in which the processes may complete.
b. If a request from process P1 arrives for (1,1,0,0), Can the request be granted immediately?
c. If a request from process P4 arrives for (0,0,2,0), Can the request be granted immediately?

Q3. Consider the following snapshot of a system:

	Allocation				Max			
	A	B	C	D	A	B	C	D
P0	3	0	1	4	5	1	1	7
P1	2	2	1	0	3	2	1	1
P2	3	1	2	1	3	3	2	1

P3	0	5	1	0	4	6	1	2
P4	4	2	1	2	6	3	2	5

Using the banker's algorithm, determine whether or not each of the following states is unsafe. If the state is safe, illustrate the order in which the processes may complete. Otherwise, illustrate why the state is unsafe.

a. Available = (0,3,0,1)
b. Available = (1,0,0,2)

Q4. Consider the following snapshot of a system:

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions using the banker's algorithm:

a. Illustrate that the system is in a safe state by demonstrating an order in which the processes may complete.
b. If a request from process P1 arrives for (1,0,2), Can the request be granted immediately?
c. If a request from process P4 arrives for (3,3,0), Can the request be granted immediately?
d. If a request from process P2 arrives for (0,2,0), Can the request be granted immediately?

Q5. Consider the following snapshot of a system:
Using Banker's algorithm

	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	3	3	2
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	0	2	0	0	2			

Is the system in a safe state?

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